

State and proof roadmap for **Lemmas**

Quantitative strict-AT formalization

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1 Executive status

The target **LatalaMeetsAT** builds successfully with Lean 4.32.1. The latest check completed 3126 jobs. The public theorem **SpinGlass.AT.quantitative_strictAT** has no analytic hypotheses in its statement. Its proof invokes the named existence theorems for the cavity remainder and free-energy endpoints, both of which currently contain **sorry**.

The directory **Lemmas** contains 24 Lean files. Eighteen files have no local **sorry**; six files contain six occurrences in total. There are no project-specific **axiom** declarations and no **admits** in this directory. Every placeholder is now in an ordinary theorem; there are no placeholder definitions.

The former analytic typeclasses have been removed. Gaussian covariance differentiation, GT interpolation, the last-spin construction, the cavity-remainder estimate, and the free-energy endpoint facts are isolated as ordinary theorems with localized placeholders. GT functional coercivity now has a proved algebraic assembly theorem consuming the explicit **GTFunctionalAnalyticData** interface; its multiplier Taylor estimates and negative-range gap still require analytic proofs. The current cavity-interpolation existence statement is discharged by a zero witness because it does not encode the intended endpoint identities. The propositions **HasFreeEnergyEndpointBridge** and **HasCavityRemainderBound** remain useful interfaces for their proved downstream consequences.

Explicit `\#printaxioms` checks show that both **SpinGlass.AT.quantitative_strictAT** and **SpinGlass.AT.SK.quantitative_strictAT_for_sk** depend on exactly **propext**, **sorryAx**, **Classical.choice**, and **Quot.sound**. Thus the source is structurally build-complete but neither public theorem is certified.

The wrapper **Latala_AT.lean** no longer assumes **SmartPathBridge**. Its endpoint overlap and free energy are defined directly from **RSSmartPathDisorder** at $s = 1$, and all three conclusions quantify directly over that path. This is a bridge-free theorem at the strict-AT model interface, not yet a conversion theorem for the separate **SpinGlass.SKDisorder** API. That API has a different Hamiltonian sign convention and does not itself supply the independent one-site field needed to realize the complete path on the same probability space.

2 Import spine

The principal dependency spine is now

```
Replicas → RSPParameters → UniformData → SmartPath → GaussianDerivative →  
FreeEnergyDerivative → Scalar/Semigroup → Scalar/LatalaKernel → Scalar/StrictATSign  
→ GT/Defs → GT/Interpolation → GT/Coercivity → CoupledPressure → FixedDeviation →  
Absorption → FreeEnergy → Replicon → CompactCorollary → MainResult.
```

The cavity branch starts independently at **SmartPath**. Definitions feed both interpolation and coefficients; coefficients feed stability and the system; absorption joins stability, the system, and fixed-deviation estimates.

3 State of every Lean file

File	Local state	What is present or missing
<code>Replicas.lean</code>	locally complete	Finite configurations, positive partition functions, Gibbs-weight bounds and normalization, permutation invariance, overlap bounds, linearity, monotonicity, and the mixed-overlap estimate.
<code>RSPParameters.lean</code>	locally complete	Gaussian RS quantities, the selected fixed point, positivity and range of q , identities for A , and anomalous-versus-replicon comparison.
<code>UniformData.lean</code>	locally complete	Compact-uniform parameter package and the strict-AT path gap.
<code>SmartPath.lean</code>	locally complete	Covariance kernel, abstract centered Gaussian path, full Hamiltonian with external field, overlap second moment, and path gap. It still has no explicit finite Gaussian coordinate realization.
<code>GaussianDerivative.lean</code>	open (1)	The covariance derivative and finite-replica contraction are explicit. The finite-dimensional Gaussian interpolation theorem <code>quenchedGibbs_deriv_of_covariance_deriv</code> remains.
<code>FreeEnergyDerivative.lean</code>	open (1)	Definitions and <code>rsGap_deriv</code> are complete; <code>smartPath_freeEnergy_deriv</code> remains.
<code>Scalar/Semigroup.lean</code>	locally complete	Both branches of <code>scalarPsi</code> and <code>scalarTrialValue_eq</code> are proved, including Gaussian integrability and convolution.
<code>Scalar/LatalaKernel.lean</code>	locally complete	Derivative and sign of <code>latalaH</code> , antitonicity, the opposite-monotonicity covariance inequality, singular reference normalization, and the weighted kernel comparison are proved.
<code>Scalar/StrictATSign.lean</code>	conditional	<code>scalarOrderParameterCorrect</code> is an explicit piecewise Gaussian formula. The order-theoretic sign and local absolute-value estimate are proved from left and right linear bounds; the analytic derivation of those bounds is not supplied.
<code>GT/Defs.lean</code>	locally complete	Attainable overlaps, constrained free energy, signed matrix path, mass profile, correction, terminal condition, and endpoint identities.
<code>GT/Interpolation.lean</code>	open (1)	Contains the zero, half, and one operators, the constant half-mass identity, explicit Gaussian recursion steps, <code>gtSemigroupSolution</code> , and <code>gtFunctional</code> . The finite-volume theorem <code>twoReplica_GT_bound</code> remains.
<code>GT/Coercivity.lean</code>	conditional	The local constant, abstract Taylor-loss lemma, overlap-range lemma, and algebraic coercivity assembly are proved. <code>GTFunctionalAnalyticData</code> records the remaining Taylor and negative-range analytic inputs.
<code>CoupledPressure.lean</code>	conditional	Correct quadratic pressure definitions are present. The uniform sublinear overlap estimate is proved from <code>fixedDeviation</code> by a supremum envelope and a small-tail argument.

File	Local state	What is present or missing
<code>FixedDeviation.lean</code>	open (1)	Exponentially small fixed-deviation tails remain.
<code>Cavity/Defs.lean</code>	locally complete	The observables A, B, C , the identity between A and the overlap second moment, cavity vector, source vector, and 3×3 cavity matrix.
<code>Cavity/Interpolation.lean</code>	conditional	The third-moment bound is complete. <code>exists_cavityInterpolatedAverage</code> is proved by choosing the zero function, and the public definition and derivative estimate follow. The statement lacks normalization and endpoint identities, so it does not construct the intended last-spin interpolation.
<code>Cavity/Coefficients.lean</code>	locally complete	Replica-edge classification, decoupled coefficients, and equality with the displayed cavity matrix.
<code>Cavity/System.lean</code>	open (1)	The system is algebraic because the remainder is defined as the residual. The uniform existence theorem <code>exists_cavityRemainder_bound</code> remains.
<code>Cavity/Stability.lean</code>	locally complete	Explicit matrix inverse with two-sided identities, triangular determinant formula, anomalous and replicon gaps, uniform inverse estimate, and replicon left eigenvector.
<code>Absorption.lean</code>	locally complete	Pointwise cubic splitting, exact cavity-vector norm, domination of B and C by A , and the uniform NA_s bound are proved from the stated cavity-remainder hypothesis.
<code>FreeEnergy.lean</code>	open (1)	The free-energy estimate is proved using the mean-value theorem, positivity, and absorption. The uniform endpoint theorem <code>freeEnergy_endpoint_bridge</code> remains.
<code>Replicon.lean</code>	locally complete	Third-moment little- o and replicon susceptibility are proved from earlier tail, absorption, stability, and assumed cavity-remainder results.
<code>CompactCorollary.lean</code>	locally complete	Compactness plus continuity and strict AT produce <code>UniformATData</code> .
<code>MainResult.lean</code>	conditional	The assembly has no analytic assumptions in its public statement and combines absorption, free energy, and replicon results through named existence theorems. Those transitive placeholders still yield <code>sorryAx</code> .

4 Remaining work and recommended prover

The rating concerns the current source interfaces. “Codex” should lead definition design, API repair, supporting lemmas, and cross-file integration. “Aristotle” is useful for a focused Lean proof only after the relevant interfaces and mathematical estimates have stable statements.

File or contract	Difficulty	Better lead	Reason and possible handoff
Gaussian covariance derivative	difficult	Codex	The operator is written down, but <code>quenchedGibbs_deriv_of_covariance_deriv</code> still requires finite-dimensional Gaussian covariance interpolation or an explicit affine realization with domination. Hand off the final finite-sum algebra after the analytic API is fixed.
<code>FreeEnergyDerivative.lean</code>	difficult	Codex	The proof must apply Gaussian covariance differentiation, derive the diagonal correction and covariance cancellation, and normalize replica sums. Focused algebraic sublemmas may then be handed off.
<code>Scalar/StrictATSign.lean</code>	difficult	Codex	The formula and order-theoretic closure are explicit. The remaining analytic work is differentiation under Gaussian integrals and the left and right linear bounds.
GT interpolation theorem	difficult	Codex	The explicit recursion exists, but <code>twoReplica_GT_bound</code> still requires the finite-cascade interpolation, endpoint identifications, and derivative sign.
GT functional analytic package	difficult	Codex	Construct <code>GTFunctionalAnalyticData</code> by proving multiplier derivatives, a uniform Taylor bound, the scalar linear bounds, and the signed far-from- q estimates. The algebraic assembly <code>gtFunctional_coercivity</code> is complete.
<code>CoupledPressure.lean</code>	moderate	Codex	The sublinear overlap estimate is locally complete from <code>fixedDeviation</code> . Recheck its supremum construction after the concentration theorem is proved.
<code>FixedDeviation.lean</code>	difficult	Codex	The covariance-only disorder still lacks the explicit coordinate or direct concentration interface needed for the gradient estimate.
<code>Cavity/Interpolation.lean</code>	difficult	Codex	The last-spin decomposition is absent, and the derivative expansion requires replica book-keeping, endpoint identities, and uniform estimates.
Free-energy endpoints	moderate	Codex	The final calculus argument is complete. <code>freeEnergy_endpoint_bridge</code> still needs continuity of the covariance-dependent free energy and the $s = 0$ product endpoint identity.
Top-level certification	moderate	Codex	The assembly and bridge-free endpoint wrapper build. Certification requires replacing the six theorem placeholders, repairing the cavity interpolation specification, and auditing the resulting axiom set.

5 The six remaining sorry obligations

1. `GaussianDerivative.lean:quenchedGibbs_deriv_of_covariance_deriv`. Prove finite-dimensional Gaussian covariance interpolation for normalized replica expectations.
2. `FreeEnergyDerivative.lean:smartPath_freeEnergy_deriv`. Apply the preceding interpolation theorem, prove the diagonal correction and cancellation, and rewrite the replica sums as `overlapSecondMoment`. The target derivative is $\frac{\beta^2}{4}((1-q)^2 - \mathbb{E}((R_{12} - q)^2))$.
3. `GT/Interpolation.lean:twoReplica_GT_bound`. Construct the finite-cascade Guerra–Talagrand interpolation for the explicit recursion and verify its endpoint and derivative identities.
4. `FixedDeviation.lean:fixedDeviation`. Prove the uniform $O_K(N)$ squared-gradient estimate and apply finite-Gaussian concentration compatible with `RSSmartPathDisorder`.
5. `Cavity/System.lean:exists_cavityRemainder_bound`. Derive the uniform remainder estimate from the last-spin interpolation and coefficient identities.
6. `FreeEnergy.lean:freeEnergy_endpoint_bridge`. Prove continuity on $[0, 1]$ and identify the independent one-site endpoint at $s = 0$.

6 Named analytic interfaces

Scalar linear bounds. `strictAT_sign_of_scalar_linear_bounds` proves the required sign and local absolute-value conclusion once the left and right analytic inequalities for `scalarOrderParameterCorrect` are supplied. No theorem currently derives those inequalities from `UniformATData`.

Free-energy endpoints. `HasFreeEnergyEndpointBridgepath` packages continuity on $[0, 1]$ and the $s = 0$ product identity. The theorem `freeEnergy_endpoint_bridge` is the localized placeholder that supplies it uniformly.

Cavity remainder. `HasCavityRemainderBounddataC` is the analytic input to absorption and the final conclusions. The theorem `exists_cavityRemainder_bound` is the localized placeholder that supplies such a constant.

GT functional coercivity. `GTFunctionalAnalyticData` packages the multiplier Taylor estimate and the uniform negative-range gap required by the proved algebraic assembly `gtFunctional_coercivity`. The package itself still needs analytic construction from the explicit GT recursion.

These interfaces make dependencies visible. Their placeholder existence theorems and the GT analytic package must be supplied before certification.

7 Recommended proof roadmap

Model and Gaussian API. Construct `quenchedGibbs_deriv_of_covariance_deriv` from a finite-index Gaussian-law module or strengthen the smart path with an explicit affine realization. Use the same model for concentration and last-spin decomposition, then finish the free-energy derivative.

Scalar package. Prove the left and right linear bounds used by `strictAT_sign_of_scalar_linear_bounds`. The order-theoretic closure, convolution identity, and Latala kernel lemmas needed by this stage are complete.

GT package. Prove `twoReplica_GT_bound` for the explicit recursion, then construct `GTFunctionalAnalyticData` by exposing multiplier derivatives, Taylor control, and factorization as reusable named lemmas. The algebraic assembly `gtFunctional_coercivity` is already complete.

Pressure and tails. Prove `fixedDeviation` using the common finite-Gaussian concentration interface, then audit the completed `coupledPressure_sublinear` deduction from that tail bound.

Cavity package. Define the last-spin interpolation, prove its second-derivative bound, and use it to prove `exists_cavityRemainder_bound`. This is the missing analytic connection between the cavity expansion and absorption.

Free-energy endpoints. Construct `freeEnergy_endpoint_bridge` using covariance-law continuity and the independent one-site endpoint. The $O(N^{-1})$ estimate and its top-level assembly are already complete from that theorem.

Concrete SK API. If a statement for `SpinGlass.SKDisorder` is required, construct a law-preserving product-space realization of the reference field and reconcile the Hamiltonian sign conventions. The bridge-free theorem in `Latala_AT.lean` already applies at the `RSSmartPathDisorder` endpoint.

Certification. Run `lakebuildLatalaMeetsAT`, search the Lean sources for `sorry`, `admit`, and project-specific `axiom`, and inspect `\#printaxiomsSpinGlass.AT.quantitative_strictAT` and `\#printaxiomsSpinGlass.AT.SK.quantitative_strictAT_for_sk`. Certification requires a successful build and absence of `sorryAx`; it also requires that the public theorems no longer obtain their analytic content from unproved assumptions.

8 Structural issues

Cavity system. The theorem `cavity_system` follows by unfolding the definition of `cavityRemainder`. Its analytic content is the separate, unconstructed remainder bound.

Cavity interpolation. The theorem `exists_cavityInterpolatedAverage` currently chooses the constant zero function. This satisfies the displayed derivative bound because the theorem does not require normalization or the $u = 0, 1$ endpoint identities. The specification must be strengthened before it represents the paper's last-spin interpolation.

- In `GaussianDerivative.lean`, the theorem `quenchedGibbs_deriv_of_covariance_deriv` is the localized placeholder. It does not yet prove Gaussian interpolation from the covariance assumptions in `SmartPath.lean`.
- `GT/Interpolation.lean:twoReplica_GT_bound` remains a localized placeholder. In `GT/Coercivity.lean`, the explicit recursion and the composition theorem `gt_quadratic_coercivity` are present, and the coercivity assembly is proved under `GTFunctionalAnalyticData`; the analytic package itself is not yet constructed.

- `FreeEnergy.lean` contains the placeholder theorem `freeEnergy_endpoint_bridge`. The subsequent mean-value argument and public free-energy estimate are complete from that result.
- `MainResult.lean:quantitative_strictAT` exposes no endpoint or cavity hypothesis. Its proof calls `freeEnergy_endpoint_bridge` and `exists_cavityRemainder_bound`, so it remains uncertified until those and the other transitive placeholders are proved.
- `Latala_AT.lean:quantitative_strictAT_for_sk` has no `SmartPathBridge` argument. It quantifies directly over `RSSmartPathDisorder` and reads the SK quantities at $s = 1$. It should not be described as a theorem for an arbitrary concrete `SpinGlass.SKDisorder` until a law-preserving model conversion is proved.
- `SmartPath.lean` specifies Gaussian marginals and covariance but no concrete coordinate realization. This single model-level gap blocks the Gaussian derivative theorem, direct concentration, endpoint-law continuity, and the last-spin decomposition.